

Linearization, transforms and solutions

R0005E - Measurement and Feedback Control

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Typical Variables:

- w : Mass flow rate, SI-Unit: kg/s
- V : Volume, SI-Unit: m^3
- q : Volume flow rate, SI-Unit: m^3/s
- h : Liquid height, SI-Unit: m
- p : Pressure, SI-Unit: N/m^2

Element laws:

- Capacitance: $C = \frac{dV}{dp}$, where V denotes the volume of the liquid. The pressure depends on the liquid height h in the vessel and is $p = \rho gh + p_a$.
- Resistance: $w = \frac{1}{R}(p_1 - p_2)^{\frac{1}{\alpha}}$, where R is the resistance of the orifice or valve. In many cases α can be approximated to 2
- Sources: These are usually pumps. Often pumps have a non-linear curve that relates the speed of the pump with the flow and difference pressure over the pump. For a given operating point, it can be said that $\Delta p = -K\omega$.

Interconnection laws:

- Continuity relation (Conservation of matter): $\dot{m} = w_{in} - w_{out}$.
- Simplified Bernoulli's principle: Total pressure is constant along a stream line, which means that the sum of dynamic and static pressure is constant, $\frac{1}{2}\rho v^2 + \rho g z + p = \text{const}$. There z denotes the raise from a reference level.
- Force of a fluid on a piston: $f = pA$, where A is the area of the piston.

Note:

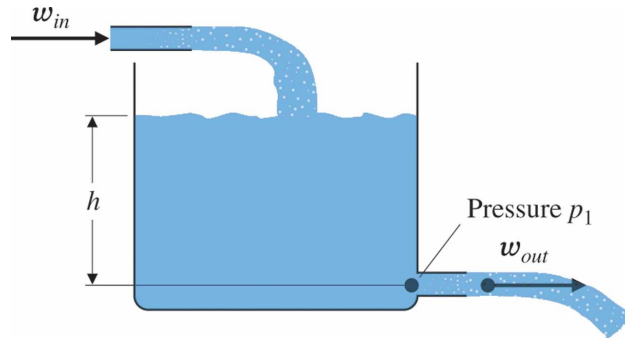
- Bernoulli's equation can be used to approximate the flow out of an open tank as

$$q = a\sqrt{2gh} \quad (1)$$

- In the (1), the friction due to the shape of the outflow is not considered. This can vary between 0.6 for sharp edges to 0.9 for smooth edges. In other words, a is multiplied by a factor which is less than 1 in real life.
- Most fluid systems are highly non-linear and do have a spatial characteristics. It is therefore inevitable to use approximations when you model such systems!

Example: Flow out of a tank

A tank is one of the most simplistic examples of a fluid system and is very common in process industry. Let's have a look at an open tank with a fixed opening at the bottom:



We will now derive a dynamic model for the tank. First we use the continuity equation for the system:

$$\dot{m} = w_{in} - w_{out} \quad (2)$$

Here, the mass flow into the tank w_{in} is the input to the model. The level h in the tank is selected as the output of the system. Therefore we need to restate (2) in terms of the level h

$$\dot{h}A\rho = w_{in} - w_{out} \quad (3)$$

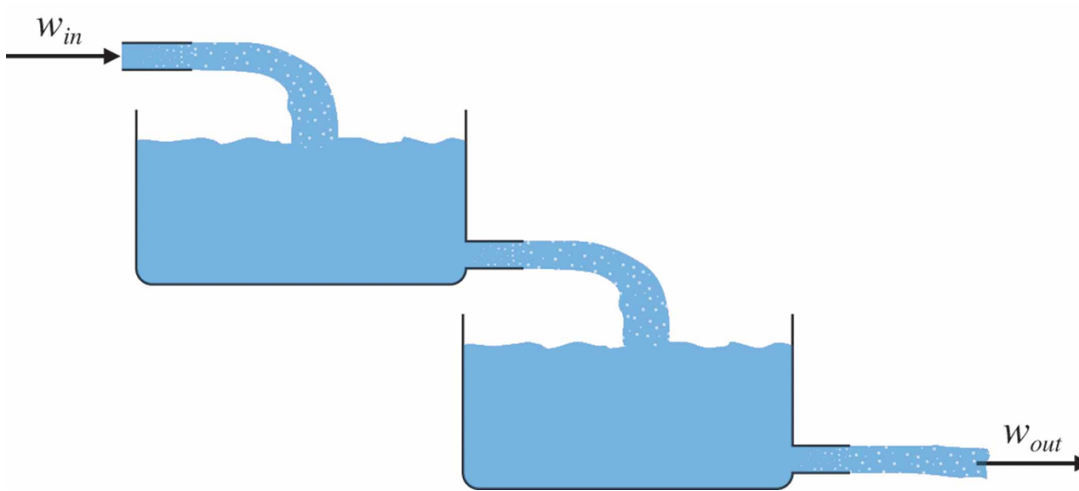
We make use of Bernoulli's equation (1) to model the flow out of the tank and restate it in input-output form get

$$\dot{h}A\rho + a\rho\sqrt{2gh} = w_{in} \quad (4)$$

Recommended problem:

Derive a dynamic model for the two-tank fluid flow system in problem 2.23.

- Determine all the variables that are of interest in the system
- What are the input and output variables?
- Derive the differential equations



Linearization

Non-linearities makes solving the differential equations and their analysis difficult, or even impossible.

Idea:

Use a linear version of the differential equation to **approximate** the non-linear behavior.

Linearity:

Two important properties:

- Superposition: $S(a + b) = S(a) + S(b)$
- Scaling: $S(\alpha a) = \alpha S(a)$

If both properties for a system are fulfilled, then the system is linear.

Remember the two examples:

- Pendulum:

$$\ddot{\theta} + \frac{g}{l} \sin \theta = \frac{T_c}{ml^2} \quad (5)$$

- Water tank:

$$\dot{h}A\rho + a\rho\sqrt{2gh} = w_{in}$$

Linearization of the pendulum

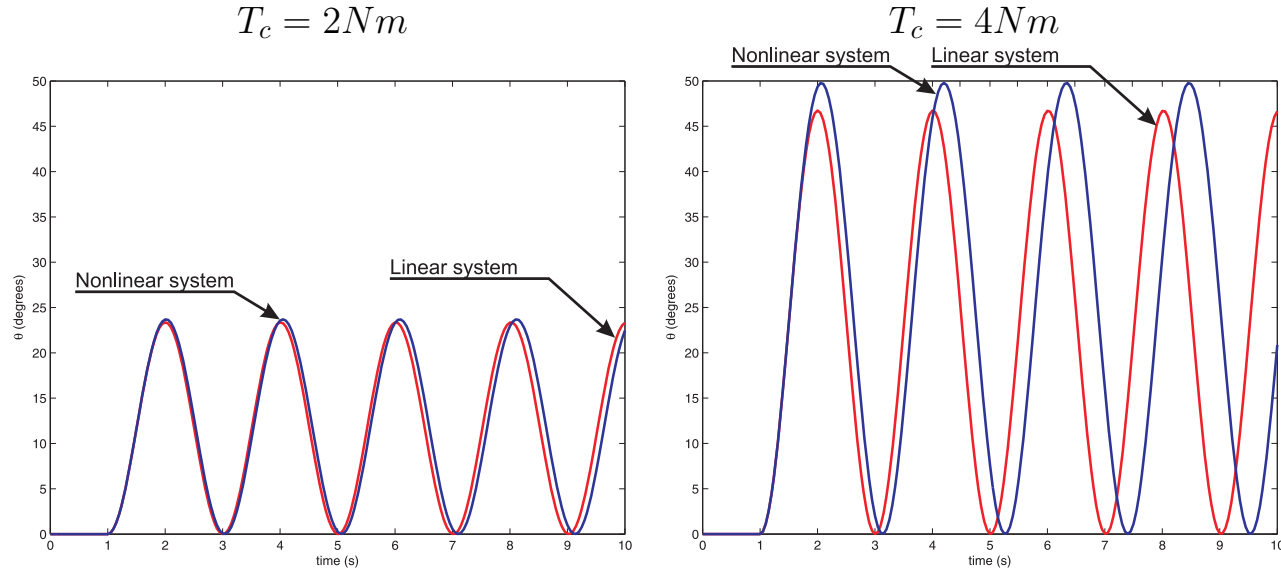
The differential equation of the pendulum has one nonlinear element: $\sin \theta$. It is known that for small angles, the following approximation holds:

$$\sin \theta \approx \theta$$

This approximation can be applied to the differential equation (5) and yields the following linear system model:

$$\ddot{\theta} + \frac{g}{l}\theta = \frac{T_c}{ml^2} \quad (6)$$

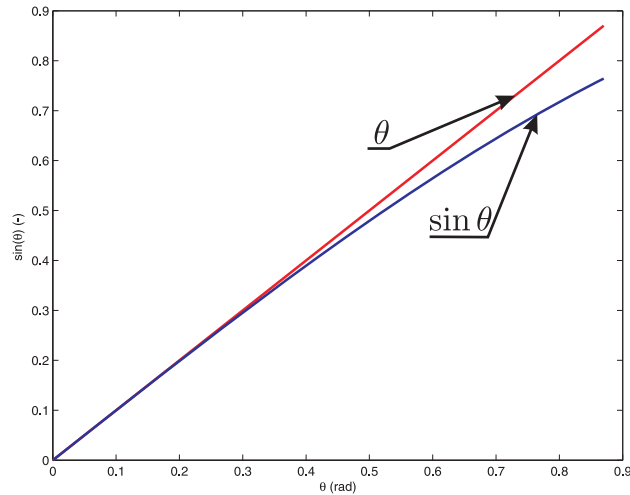
We simulate the model now with the same parameter settings as before with two different step-like input signals for the torque T_c :



Some observations:

- If θ is less than 15° then the approximation seem to work well.
- For large angles, the linear system underestimates the angles and reacts faster than the non-linear system.
- The nonlinear system behaves like a linear system for θ close to zero.

Error:



It is obvious that $\sin \theta \leq \theta$, which means that $\ddot{\theta}$ is not reduced as much in the nonlinear system as in the linear system.

Linearization of the water tank

For the pendulum it was quite natural that an approximation of $\sin \theta$ close to zero is a good choice. Obviously, the pendulum will settle at that angle. This is **rarely** the case!

Now we want to find a linear system that represents the water tank. In the water tank, the outflow depends on the level in the tank, which in turn depends on the inflow:

$$\dot{h}A\rho + a\rho\sqrt{2gh} = w_{in}$$

Some observations:

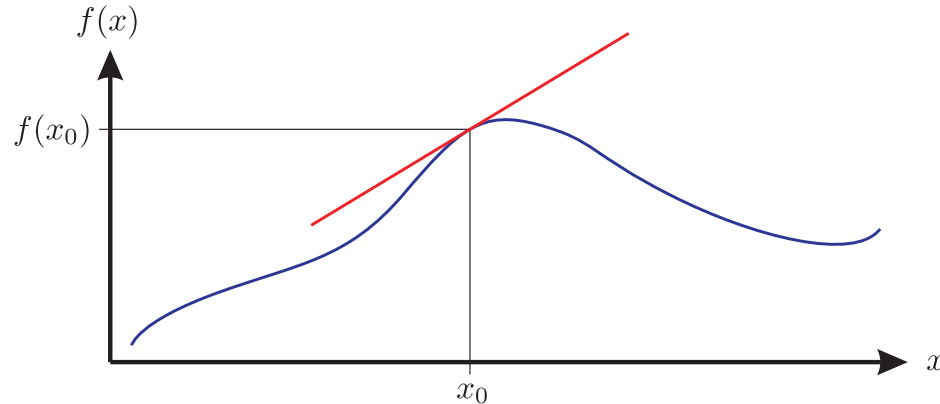
- The level in the water tank will settle at different levels depending on the average inflow
- The higher the average inflow level, the higher the level in the tank on the average.
- There are different operating points depending on the average inflow.

Idea:

- Determine an operating point for the level h . A good choice is usually an equilibrium point!
- Derive a linear function that approximates the square root function.
- Replace the nonlinear function with the linear one.

General approach

Assume a function $f(x)$ is given as follows:



Now we want to find the linear approximation of that function for the operating point x_0 .

The function $f(x)$ can be described in terms of the operating point:

$$\begin{aligned}x(t) &= x_0 + \Delta x(t) \\ f(x(t)) &= \underbrace{f(x_0)}_{=f_0} + \Delta f(x(t))\end{aligned}$$

There x_0 and f_0 are the nominal values and $\Delta x(t)$ and $\Delta f(x(t))$ are the incremental values.

Close to the operating point (x_0, f_0) the function value can be expressed by a linear relationship:

$$f(x(t)) = f_0 + K(x(t) - x_0), \quad \text{with } K = \left. \frac{df}{dx} \right|_{x_0} \quad (7)$$

We can rewrite (7) as

$$f(x(t)) - f_0 = K(x(t) - x_0) \quad (8)$$

$$\Rightarrow \Delta f(x(t)) = K \Delta x(t) \quad (9)$$

Now we have found our linear function, which is valid close to the operating point (x_0, f_0) . Be sure to remember that this is an approximation.

Generally, this is the truncated Taylor series expansion after the first two terms:

$$f(x) = f(x_0) + \left. \frac{df}{dx} \right|_{x_0} (x - x_0) + \frac{1}{2!} \left. \frac{d^2f}{dx^2} \right|_{x_0} (x - x_0)^2 + \dots$$

Note:

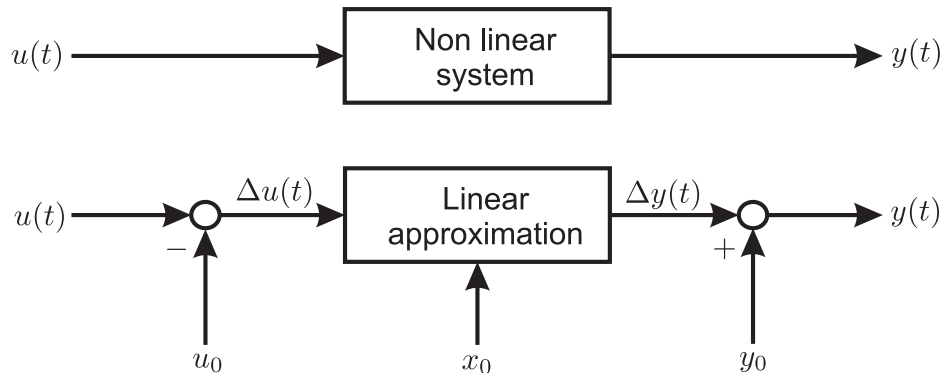
- Using a set of operating points a complex nonlinearity can be represented by a set of linear relationships.
- The operating point is usually a point, where the system is at rest (equilibrium). Thus, all derivatives are zero.
- Using the nominal value for the input and setting all derivatives to zero yields the nominal state and output values.

Procedure:

1. Determine the operating point for the input condition.
2. Write all terms as linearized terms using nominal and incremental variables. Use the Taylor series expansion where needed.
3. Rewrite the equations so that only incremental variables are used.
4. Determine the initial conditions for all incremental variables.

Important note:

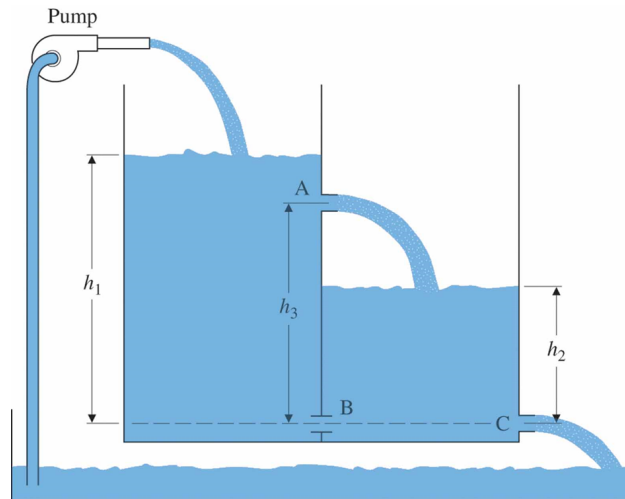
In order to calculate the correct states and output values, it is necessary to make use of the nominal values and the initial conditions for both states and outputs.



Recommended problem:

Derive a dynamic model for the system in problem 2.24.

- Derive the differential equations
- Perform a linearization
- Derive the transfer function for the linearized system.



Additional material

You can find many videos on linearization. All bring in a different perspective and usually derive the linearization from different perspectives.

- Blending Process: Linearization Example

<https://www.youtube.com/watch?v=pQa3aGfd0e8>

Here you see the linearization of a process with several variables

- Intro to Control - 5.2 System Linearization

<https://www.youtube.com/watch?v=qBFWbi0JPBA>

Here you get an understanding on how the operating point conditions cancel out.

Superposition

Remember the superposition property of linear systems:

- Input $u_1(t)$ gives output $y_1(t)$.
- Input $u_2(t)$ gives output $y_2(t)$.
- Then, $au_1(t) + bu_2(t)$ gives output $ay_1(t) + by_2(t)$.

As a result we can do the following:

Split the input signal $u(t)$ into a sum of input signals $\mu_i(t)$ for which we know the output signals. Then, we know the output signal $y(t)$ and that for any input signal $u(t)$.

Example:

Input signal $\mu(t)$ gives the output signal $\nu(t)$. Compose the input signal as

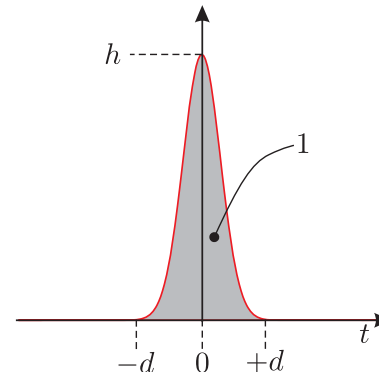
$$u(t) = \sum_i a_i \mu(t - \tau_i) \quad (10)$$

$$\Rightarrow y(t) = \sum_i a_i \nu(t - \tau_i) \quad (11)$$

This means that there is a set of (a_i, τ_i) which describes the system response.

What are good $\mu(t)$? – Unit impulse!

$$\delta(t) = 0, \quad t \neq 0$$
$$\int_a^b \delta(t) dt = 1, \quad a < 0 < b$$



Now we can apply the unit impulse on an arbitrary signal $f(t)$:

$$\int_{-d}^{+d} f(t)\delta(t)dt = f(0)$$

This means that the function $f(t)$ is sampled at $t = 0$. In order to sample the function $f(t)$ at a different time t_i we can do the following:

$$\int_a^b f(t)\delta(t - t_i)dt = f(t_i)$$

Instead of sampling or decomposing the signal, $\delta(t)$ can also be used to reconstruct or compose a signal, when all the samples are known:

$$u(t) = \int_{-\infty}^{\infty} u(\tau)\delta(t - \tau)d\tau \quad (12)$$

If we know the output of a system for the input $\delta(t)$, then we can reconstruct the output signal. The output of a system for a given input is denoted **response**. Thus, the output of the system to a unit impulse input is called the **impulse response** of the system and will be denoted $h(t)$.

$$\Rightarrow y(t) = \int_{-\infty}^{\infty} u(\tau)h(t - \tau)d\tau \quad (13)$$

This integral is the convolution of $u(t)$ with $h(t)$, and will be denoted $(u * h)(t)$ in the sequel.

Transfer function

Assume we choose the input signal $u(t) = e^{st}$ and then calculate the output $y(t)$:

$$\begin{aligned}\stackrel{(12)}{\implies} y(t) &= \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau \\ &= \int_{-\infty}^{\infty} h(\tau) e^{st} e^{-s\tau} d\tau \\ &= e^{st} \underbrace{\int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau}_{\triangleq H(s)} \\ &= e^{st} H(s)\end{aligned}$$

$y(t)$ is the input scaled by $H(s)$. Thus, $H(s)$ is the **gain** between input and output. $H(s)$ is the **transfer function** of the system described by the impulse response $h(t)$. The transfer function is the Laplace transform of the impulse response.

The integral that derives $H(s)$ is the double sided Laplace transform. In this form it considers all history, as it starts from $-\infty$.

One-sided Laplace transform:

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt, \quad s = \sigma + j\omega$$

Obviously, we use $t = 0$ as the starting time. At this time there can be initial conditions that affect the behavior of the system.

Convolution and the Laplace transform

Assume:

- We know the impulse response $h(t)$ of a linear system,
- and

$$u(t) = \begin{cases} \text{something} & t \geq 0 \\ 0 & t < 0 \end{cases} \quad (14)$$

Determine the Laplace transform of the output $y(t)$

We will start from the definition of the laplace transform of $y(t)$:

$$\begin{aligned} Y(s) &= \int_0^\infty y(t) e^{-st} dt \\ &= \int_0^\infty (u * h)(t) e^{-st} dt \\ &= \int_0^\infty \int_{-\infty}^\infty u(\tau) h(t - \tau) d\tau e^{-st} dt \\ &\stackrel{(14)}{=} \int_0^\infty \int_0^\infty u(\tau) h(t - \tau) d\tau e^{-st} dt \\ &= \int_0^\infty \int_0^\infty u(\tau) h(t - \tau) e^{-st} dt d\tau \end{aligned}$$

Now we can introduce the following substitution: $v = t - \tau \Rightarrow t = v + \tau \Rightarrow dv = dt$.

This results in:

$$Y(s) = \int_0^\infty \int_{-\tau}^\infty u(\tau) h(v) e^{-sv} e^{-s\tau} dv d\tau$$

Assumption: $h(t) = 0, \quad t < 0$.

This assumption is called causality. There can not be a response before an input is present. Using this assumption the second integral can be modified and we get

$$Y(s) = \int_0^\infty \int_0^\infty u(\tau) h(v) e^{-sv} e^{-s\tau} dv d\tau$$

The double integral can now be rearranged

$$\begin{aligned} Y(s) &= \int_0^\infty h(v) e^{-sv} dv \int_0^\infty u(\tau) e^{-s\tau} d\tau \\ &= H(s) \cdot U(s) \end{aligned}$$

So, the convolution resolves into a multiplication in the Laplace domain :-)

Properties of the Laplace transform

We will have a look at some selected properties of the Laplace transform:

	Time domain	Laplace domain
1. Superposition:	$\alpha f_1(t) + \beta f_2(t)$	$\alpha F_1(s) + \beta F_2(s)$
2. Time delay:	$f(t - \lambda)$	$F(s)e^{-s\lambda}$
3. Time product:	$f_1(t) \cdot f_2(t)$	$(F_1 * F_2)(s)$
4. Frequency shift:	$f(t)e^{-at}$	$F(s + a)$
5. Multiplication by time:	$t \cdot f(t)$	$-\frac{d}{ds}F(s)$
6. Convolution:	$(f_1 * f_2)(t)$	$(F_1 \cdot F_2)(s)$

We will have a closer look at the differentiation and integration property of the Laplace transform.

7. Differentiation:

$$\begin{aligned}\mathcal{L}\left(\frac{d}{dt}f(t)\right) &= \int_0^{\infty} \underbrace{\frac{d}{dt}f(t)}_{u'} \underbrace{e^{-st}}_v dt, & \frac{d}{dt}(uv) &= \frac{du}{dt}v + u\frac{dv}{dt} \\ &= [e^{-st}f(t)]_0^{\infty} - \left(-s \int_0^{\infty} f(t)e^{-st}dt\right) \\ &= 0 \cdot f(\infty) - 1 \cdot f(0_-) + s \cdot F(s) \\ &= sF(s) - f(0_-)\end{aligned}$$

Here, $f(0_-)$ is the initial value of f at time $t = 0$, seen from $-\infty$.

Now we can derive the n^{th} derivative by applying the differentiation property over and over. For the second derivative this yields:

$$\mathcal{L}\left(\frac{d^2}{dt^2}f(t)\right) = s(sF(s) - f(0_-)) - f'(0_-)$$

8. Integration:

The integration is the inverse of the differentiation and yields

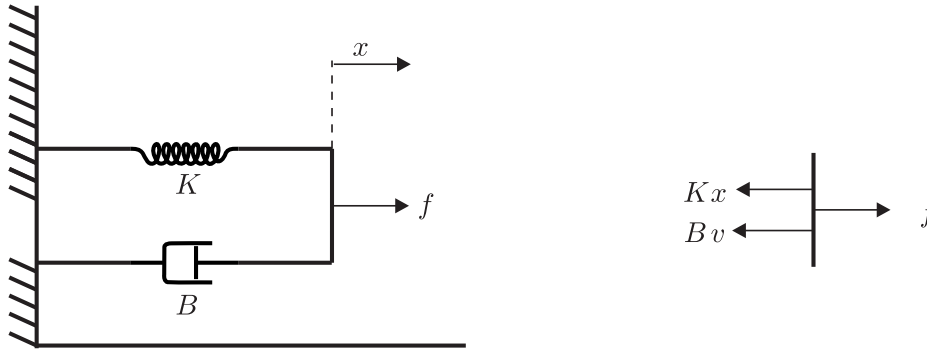
$$\mathcal{L}\left(\int_0^t f(\tau)d\tau\right) = \frac{1}{s}F(s)$$

Recommended problem: 3.3

Transfer functions from differential equations

Until now we have derived the transfer function from the impulse response. But, the impulse response might not be available. Instead, the linear differential equation of a system might be available.

Let's have a look at the following simple example:



From the free body diagram we get:

$$B\dot{x} + Kx = f(t)$$

Now we apply the Laplace transform to the differential equation:

$$\begin{aligned} B\mathcal{L}(\dot{x}(t)) + K\mathcal{L}(x(t)) &= \mathcal{L}(f(t)) \\ \Rightarrow B(sX(s) - x(0)) + KX(s) &= F(s) \\ \Rightarrow X(s) &= \frac{1}{Bs + K} (Bx(0) + F(s)) \end{aligned}$$

Note: Remember the trick that was used to derive the input-output form.

Clearly, there are two parts: One that depends on the initial condition and the other that depends on the input.

We can now rewrite the equation into the following form:

$$X(s) = \underbrace{\frac{x(0)}{s + K/B}}_{\text{Zero input response}} + \underbrace{\frac{F(s)/B}{s + K/B}}_{\text{Zero state response}}$$

Now we can transform $X(s)$ back to the time domain and the complete solution for the differential equation including initial conditions $x(0)$ and input signal $f(t)$ is found.

Some typical input signals:

	Time domain	Laplace domain
1. Dirac (impulse):	$\delta(t)$	1
2. Step function:	$u(t) = \begin{cases} A, & t \geq 0 \\ 0, & t < 0 \end{cases}$	$\frac{A}{s}$
3. Ramp function:	$u(t) = \begin{cases} Kt, & t \geq 0 \\ 0, & t < 0 \end{cases}$	$\frac{K}{s^2}$
4. Sine wave function:	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
5. Cosine wave function:	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$

The way back ...

Let's check for the output $x(t)$ for $f(t) = \delta(t)$ and $x(0) = 0$.

$$X(s) = \frac{1/B}{s + K/B} \quad \Rightarrow \quad x(t) = \frac{1}{B} e^{-\frac{K}{B}t}$$

In this case $x(t) = h(t)$, which is the impulse response. Clearly, the parameter K/B determines the behavior of the response:

- $\frac{K}{B} \rightarrow 0$: The time to reach a steady state tends to infinity.
- $\frac{K}{B} < 0$: Then the response $x(t)$ tends to infinity as $t \rightarrow \infty$. It is unstable.

How does the step response look like with $x(0) = 0$? In this case, the input signal $f(t)$ becomes:

$$f(t) = \begin{cases} A, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

The response then results in

$$X(s) = \frac{1/B}{s(s + K/B)} \quad \Rightarrow \quad x(t) = \int_0^t \frac{1}{B} e^{-\frac{K}{B}\tau} d\tau = \frac{1}{K} \left(1 - e^{-\frac{K}{B}t}\right)$$

Note: In both cases the parameter $\frac{K}{B}$ is a system constant. It determines the "fastness" of the response

$\Rightarrow$ It is a time constant!

Transient response and steady state

As we could see in the last example, a system will respond on an input and if the output of the system is **bounded** and the system is **stable**, then it will reach a final value.

This final value is called the **steady state value**!

Definition:

- Transient response: As long as the derivatives in a system are non-zero, then the transient response is non-zero. The transient response decays to zero, when the steady state is reached.
- Steady state response: When the system has come to a rest and all derivatives have decayed to zero then the steady state response is remaining.

Final and initial value theorem

There are two important theorems, which are very useful for the analysis of system (with and without controller).

1. Final value theorem:

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$$

2. Initial value theorem:

$$\lim_{t \rightarrow 0} y(t) = \lim_{s \rightarrow \infty} sY(s)$$

Important note: The final value theorems is only valid if the limit of $y(t)$ is bounded and defined for $t \rightarrow \infty$!

Example:

$$X(s) = \frac{1/B}{s(s + K/B)}$$
$$\lim_{s \rightarrow 0} sX(s) = \frac{1/B}{s + K/B} = \frac{1}{K}$$

Important conclusion:

All properties of the system can be derived from the analysis of the transfer function. We do not need to solve the differential equation!

Solution of the differential equation

In case we need to get back to the time and domain and thus, compute the solution then we proceed as follows:

Procedure:

1. Derive the transform of the differential equation with all initial conditions and for $t > 0$.
2. Solve the algebraic equation.
3. Evaluate the inverse transform (Table look-up, e.g. Table A2)

If the inverse transform is not available, then we have to manually transform back. This can be done by using the partial fraction expansion.

Partial Fraction Expansion

A transfer function is usually a fraction. If we can decompose the fraction into a sum of fractions where the inverse transform of the addends is known then the problem is solved.

There are 3 cases:

1. Distinct poles:

$$F(s) = \frac{1}{(s + a_1)(s + a_2)(s + a_3) \cdots} = \frac{A_1}{s + a_1} + \frac{A_2}{s + a_2} + \frac{A_3}{s + a_3} + \cdots \quad (15)$$

2. Repeated poles:

$$\begin{aligned} F(s) &= \frac{1}{(s + a_1)^k (s + a_2)(s + a_3) \cdots} \\ &= \frac{A_{11}}{s + a_1} + \frac{A_{12}}{(s + a_1)^2} + \cdots + \frac{A_{1n}}{(s + a_1)^k} + \frac{A_2}{s + a_2} + \frac{A_3}{s + a_3} + \cdots \end{aligned} \quad (16)$$

3. Complex poles:

$$\begin{aligned} F(s) &= \frac{1}{(s^2 + a_{11}s + a_{10})(s + a_2) \cdots} \\ &= \frac{A_{11}s + A_{12}}{s^2 + a_{11}s + a_{10}} + \frac{A_2}{s + a_2} + \cdots \end{aligned}$$

Do not split the pole pair!

General Procedure:

1. Bring the sum of fractions on a common denominator.
2. Compare the coefficients of the numerator for each s^n .
3. Solve the set of algebraic equations.
4. Transform each of the addends in the partial fraction expansion back to the time domain.

Cover-up method:

In the cover-up method, coefficients for distinct poles and repeated poles can be found directly. This reduces the amount of coefficients to be computed in step 2, largely.

For the coefficients A_i in (15), it can be shown

$$A_i = (s + a_i)F(s)|_{s=-a_i}$$

Similarly, the coefficients A_{ij} in (16) for poles with multiplicity k can be computed as:

$$A_{i(k-j)} = \frac{1}{j!} \left[\frac{d^j}{ds^j} (s + a_i)^k F(s) \right]_{s=-a_i}, \quad j = 0, \dots, k-1.$$

Recommended problems

The problems in the text book help you to get familiar with the mathematics and then apply it to a rather realistic system.

- Problem 3.7: Practice the partial fraction expansion and the inverse transform
- Problem 3.9: Practice the solving of initial value problems
- Problem 3.16: Formally derive the transfer function for the tank system in figure 2.55

Additional material

In relation to this lecture the following videos are of interest:

- Control Systems Lectures - LTI Systems
https://www.youtube.com/watch?v=3eDDTFcSC_Y
- Control Systems Lectures - Transfer Functions
<https://www.youtube.com/watch?v=RJ1eGwXorUk&t=8s>
- Final Value Theorem and Steady State Error
<https://www.youtube.com/watch?v=PXxveGoNRUw>
Here the first part of the video is of interest for the final value theorem. The later part which is about the steady state error will be discussed in a later lecture. System types are not discussed in the course, but if you are interested you can of course have a look and discuss with us if you have questions.

For a repetition or introduction to the Fourier and Laplace transforms the following two videos are of interest. It is good to watch them in the given order.

- Control Systems Lectures - Time and Frequency Domain
https://www.youtube.com/watch?v=noycLIZbK_k
- The Laplace Transform - A Graphical Approach
<https://www.youtube.com/watch?v=ZGPtPkTft8g>